MINI INVENTORY MANAGEMENT SYSTEM

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PROJECT WORK

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BONAFIDE CERTIFICATE

Certified that this Report titled "**MINI INVENTORY MANGEMENT SYSTEM "** for the course **U23CS382–Python Programming** is the bonafide work of

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Todeclarethattheproject entitled **"MINI INVENTORY MANGEMENT SYSTEM "** submitted in partial fulfillment to the University as the projectwork of Bachelor of Engineering (Computer and Communication Engineering) Degree ,is a record of original work done by me under the supervision and guidance of **Dr. C. Ganesh , M.E., Ph.D.** Associate Professor, Department of Computer and Communication Engineering, Sri Eshwar College of Engineering, Coimbatore.

Place: Coimbatore
Date:

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TABLEOFCONTENTS

ABSTRACT

LIST OF FIGURES

1. INTRODUCTION..... 2

2. SYSTEMANALYSISANDDESIGN 3

1. EXISTINGSCENARIO..... 3

2. PROBLEMSTATEMENT 4

3. PROPOSEDSOLUTION 5

1. OVERVIEW 5

2. BLOCK DIAGRAM FOR MINI INVENTORY MANAGEMENT SYSTEM..... 6

3. DEVELOPMENT PHASES.....7

4 IMPLEMENTATIONSANDRESULTS 11

1. CODE 11

2. OUTPUT 26

5 CONCLUSION 29

6 REFERENCES 30

LIST OF FIGURES

Figure No.	Figure Name	Page No.
2.2.1	Block Diagram for Mini inventory management system	6
3.2.1	Evaluation Result	26
3.2.2	Accuracy/Response time graph	27
3.2.3	Web Output	27
3.2.4	LawBot Response	28

ABSTRACT

The Mini Inventory Management System is a software application developed to efficiently manage and track inventory operations in small-scale businesses or organizations.

The primary objective of this system is to simplify the process of maintaining stock details, monitoring product availability, and reducing manual errors associated with traditional record-keeping methods.

This system provides functionalities such as adding, updating, and deleting product records, tracking stock levels, and generating basic reports.

It enables users to maintain accurate information about inventory items, including product name, quantity, price, and supplier details.

The system also helps in identifying low-stock items, thereby assisting in timely restocking and preventing shortages.

The Mini Inventory Management System is designed with a user-friendly interface and can be implemented using programming languages such as Python, Java, or C++, along with a database for data storage. It improves efficiency, saves time, and ensures better organization of inventory data.

Overall, this project demonstrates how a simple automated system can replace manual inventory management, leading to increased accuracy, reliability, and productivity in business operations.

OBJECTIVE

The primary objective of the Mini Inventory Management System is to design and develop a simple, efficient, and user-friendly application that helps in managing inventory operations effectively. The system aims to eliminate the drawbacks of manual record-keeping by reducing human errors, data redundancy, and time consumption. It focuses on maintaining accurate and up-to-date records of all inventory items, including details such as product name, quantity, price, and supplier information.

Furthermore, the system is intended to provide essential functionalities like adding new products, updating existing records, and deleting outdated information, thereby ensuring smooth data management. It also helps in continuously monitoring stock levels and identifying low-stock or out-of-stock items, enabling timely restocking and preventing business interruptions.

Another important objective is to improve decision-making by generating basic reports related to inventory status, stock availability, and product movement. The system is designed to enhance efficiency, productivity, and reliability in inventory handling while saving time and effort. Additionally, it aims to offer a simple interface that can be easily understood and operated by users with minimal technical knowledge, making it suitable for small-scale businesses and organizations.

1.EXISTINGSCENARIO

In many small-scale businesses and organizations, inventory management is still carried out using manual methods such as maintaining records in notebooks, spreadsheets, or basic files. In this existing scenario, all details related to products, stock levels, and transactions are recorded manually, which makes the process time-consuming and prone to human errors. Updating inventory records requires significant effort, and there is a high chance of mistakes such as incorrect data entry, duplication, or loss of information.

Additionally, tracking stock levels becomes difficult, especially when dealing with a large number of items. There is no proper system to alert users about low-stock or out-of-stock products, which can lead to delays in restocking and affect business operations. Generating reports and analyzing inventory data is also challenging, as it requires manual calculations and verification.

Furthermore, the lack of a centralized system results in poor data organization and limited accessibility. Only one person can update records at a time, and retrieving specific information can be slow and inefficient. Overall, the existing manual inventory management system lacks accuracy, reliability, and efficiency, highlighting the need for an automated solution.

2.PROBLEMSTATEMENT

In many small and medium-scale businesses, inventory management is still performed using traditional methods such as handwritten records, ledgers, or simple spreadsheet applications. These methods are inefficient, time-consuming, and highly susceptible to human errors such as incorrect data entry, duplication, and data loss. As the volume of inventory increases, managing and updating records manually becomes increasingly difficult and unorganized.

One of the major problems in the existing system is the lack of real-time tracking of inventory. Businesses often face issues such as overstocking or understocking due to inaccurate or outdated information.

The absence of an automated alert system makes it difficult to identify low-stock or out-of-stock items, which can lead to delays in restocking and disruption of daily operations. Moreover, there is no centralized database to store and manage inventory data, resulting in poor data accessibility and lack of coordination.

Retrieving specific information or generating reports requires significant time and effort, making decision-making slow and less effective. Security of data is also a concern, as manual records can be easily lost or damaged.

Therefore, these challenges highlight the need for a Mini Inventory Management System that can automate inventory processes, maintain accurate and up-to-date records, provide quick access to information, and improve overall efficiency, reliability, and productivity in inventory management

.

2.1 OVERVIEW

The Mini Inventory Management System is a simple software application designed to help small businesses and organizations efficiently manage their inventory operations. It provides a structured and automated way of handling stock-related activities such as storing product details, updating quantities, and tracking item availability. The system replaces traditional manual methods with a digital approach, ensuring better accuracy and organization of data.

This system allows users to perform key operations such as adding new products, modifying existing records, and deleting outdated entries. It maintains important information including product name, quantity, price, and supplier details in a centralized database, making it easy to access and manage. Additionally, it helps monitor stock levels and supports timely decision-making by identifying items that need restocking.

The Mini Inventory Management System is designed with a user-friendly interface, making it suitable for users with minimal technical knowledge. It can be developed using programming languages like Python, Java, or C++ along with a database management system for data storage. By automating routine inventory tasks, the system improves efficiency, reduces errors, saves time, and enhances overall productivity.

Overall, this project demonstrates how a basic automated solution can simplify inventory management processes and provide a reliable and effective tool for small-scale inventory control.

2.2 BLOCK DIAGRAM

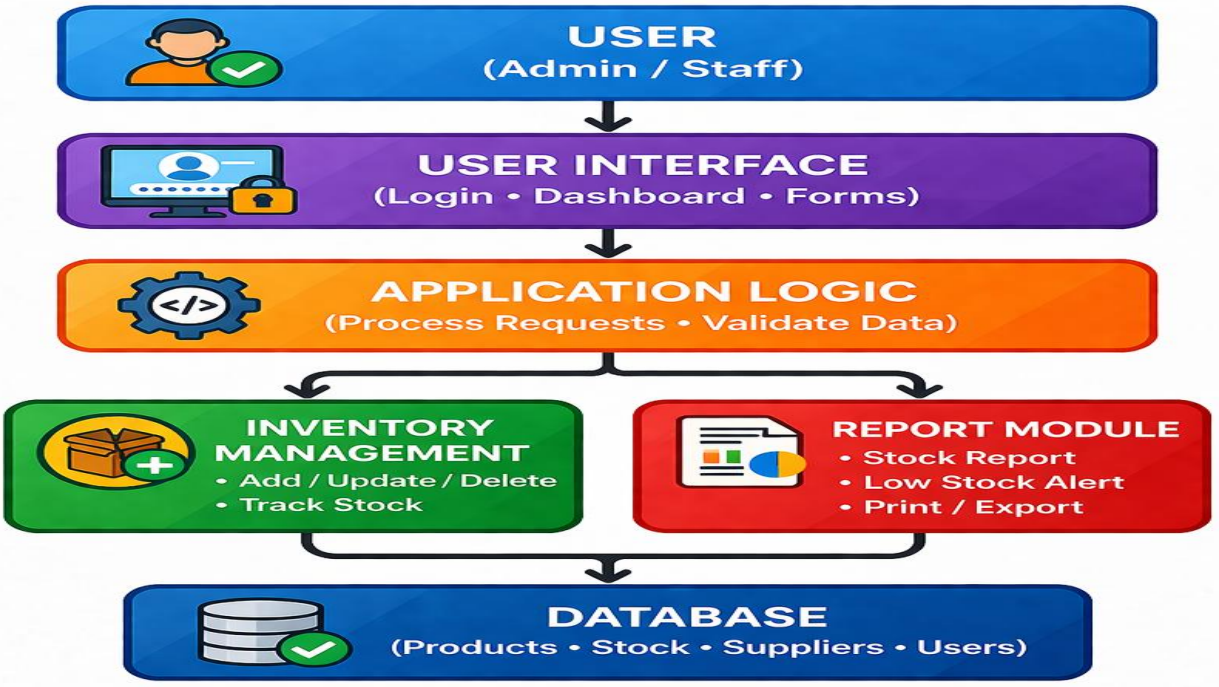

Fig 2.2.1 BLOCK DIAGRAM FOR MINI INVENTORY MANAGEMENT SYSTEM

2.2 DEVELOPMENT PHASE

PHASE 1:Planning & Requirement Analysis

This is the first and most important phase of the system development process. In this stage, the problem is clearly identified and analyzed. Information is gathered from users such as shopkeepers, staff, or administrators to understand how the current inventory system works and what difficulties they face. Based on this, detailed requirements are collected, including **functional requirements** (like adding products, updating stock, searching items, generating reports) and **non-functional requirements** (such as performance, security, and user-friendliness).

Additionally, a **feasibility study** is conducted to determine whether the project is practical. This includes technical feasibility (availability of tools and technology), economic feasibility (cost and budget), and operational feasibility (ease of use and acceptance by users). This phase sets a strong foundation for the entire project.

PHASE 2:System Design

In the system design phase, the overall structure and architecture of the system are planned. Based on the requirements gathered earlier, the system is divided into different modules such as **User Interface, Application Logic, Inventory Management, Report Module, and Database**.

Design tools such as **block diagrams, data flow diagrams (DFD), and entity-relationship (ER) diagrams** are created to visually represent how data flows within the system. The database structure is also designed by defining tables, fields, primary keys, and relationships between different entities. This phase acts as a blueprint that guides developers during the coding process.

PHASE 3:Development

This phase involves the actual implementation of the system. Developers write code using programming languages such as Python, Java, or C++. Each module designed in the previous phase is developed step by step. Features such as adding new products, updating stock levels, deleting records, and generating reports are implemented.

The database is created using tools like MySQL or SQLite, and proper connections are established between the application and the database. Developers also ensure proper validation of input data to avoid errors. This phase transforms the design into a working system.

Phase 4: Testing

In the testing phase, the developed system is thoroughly checked to ensure it is free from errors and works according to requirements. Various types of testing are performed, including **unit testing** (testing individual components), **integration testing** (testing interaction between modules), and **system testing** (testing the entire system as a whole).

Errors such as incorrect calculations, data mismatches, or system crashes are identified and corrected. Testing ensures that the system is reliable, accurate, and performs efficiently under different conditions. This phase is crucial to deliver a quality product

Phase 5: Implementation (Deployment)

After successful testing, the system is deployed in the real environment where it will be used. The software is installed on user systems, and necessary configurations are completed. Users such as staff or administrators are trained on how to use the system effectively.
Initial support is provided to handle any issues faced during the first usage. Data from old systems (if any) may also be transferred to the new system. This phase ensures that the system is successfully introduced and used in real-world conditions.

PHASE 6:Maintenance

Maintenance phase is the final and continuous stage of the Mini Inventory Management System, where the system is monitored, updated, and improved after deployment. This phase ensures that the system continues to function effectively in a real-world environment and adapts to changing user requirements over time.

Maintenance involves several important activities. One of the key aspects is **corrective maintenance**, where errors or bugs that were not identified during the testing phase are detected and fixed. These issues may arise during actual usage, and timely correction is essential to maintain system reliability.

Another important activity is **adaptive maintenance**, which involves modifying the system to accommodate changes in the environment, such as updates in operating systems, hardware, or database technologies. This ensures that the system remains compatible and functional even as technology evolves.

Perfective maintenance focuses on enhancing the system by adding new features or improving existing functionalities based on user feedback. For example, new report formats, better search options, or improved user interface design may be introduced to increase efficiency and user satisfaction.

Additionally, **preventive maintenance** is carried out to improve system performance and prevent future problems. This may include optimizing database performance, improving code efficiency, and regularly backing up data to avoid data loss.

Regular monitoring, updates, and user support are essential parts of this phase. Proper documentation is also maintained to keep track of changes and improvements. Overall, the maintenance phase plays a crucial role in ensuring the long-term success, reliability, and usability of the Mini Inventory Management System.

1. **Code**
2. import json

```
FILE_NAME = "inventory.json“

 def load_inventory():
  try:
 with open(FILE_NAME, "r") as file:
 return json.load(file)
  except:
   return {
def save_inventory(inventory):
with open(FILE_NAME, "w") as file:
 json.dump(inventory, file, indent=4)


def add_product(inventory):
product_id = input("Enter Product ID: ")

if product_id in inventory:
print("Product already exists!\n")
return

name = input("Enter Product Name: ")
quantity = int(input("Enter Quantity: "))
price = float(input("Enter Price: "))

inventory[product_id] = {
"name": name,
"quantity": quantity,
"price": price
}

save_inventory(inventory)
print("Product added successfully!\n")
```

```
def view_products(inventory):
if not inventory:
print("Inventory is empty.\n")
return

print("\n--- Inventory ---")
for pid, details in inventory.items():
print(f"ID: {pid}")
print(f"Name: {details['name']}")
print(f"Quantity: {details['quantity']}")
print(f"Price: {details['price']}")
print("---------------------")
print()


def update_product(inventory):
product_id = input("Enter Product ID to update: ")

if product_id in inventory:
quantity = int(input("Enter new Quantity: "))
price = float(input("Enter new Price: "))

inventory[product_id]["quantity"] = quantity
inventory[product_id]["price"] = price

save_inventory(inventory)
print("Product updated successfully!\n")
else:
print("Product not found!\n")
```

```
def delete_product(inventory):
product_id = input("Enter Product ID to delete: ")

if product_id in inventory:
del inventory[product_id]
save_inventory(inventory)
print("Product deleted successfully!\n")
else:
print("Product not found!\”)

def search_product(inventory):
product_id = input("Enter Product ID to search: ")

if product_id in inventory:
details = inventory[product_id]
print("\nProduct Found:")
print(f"Name: {details['name']}")
print(f"Quantity: {details['quantity']}")
print(f"Price: {details['price']}\n")
else:
print("Product not found!\n")
```

```
def menu():
inventory = load_inventory()

while True:
print("==== Inventory Management System ====")
print("1. Add Product")
print("2. View Products")
print("3. Update Product")
print("4. Delete Product")
print("5. Search Product")
print("6. Exit")

choice = input("Enter your choice: ")

if choice == '1':
add_product(inventory)
elif choice == '2':
view_products(inventory)
elif choice == '3':
update_product(inventory)
elif choice == '4':
delete_product(inventory)
elif choice == '5':
search_product(inventory)
elif choice == '6':
print("Exiting program...")
break
else:
print("Invalid choice! Try again.\n")

menu()
```

(+/-

Invent ory Management System

1. Add Product
2. View Products
3. Update Product
4. Delete Product
5. Search Product
6. Exit

Enter your choice: 1

Enter Product ID: 101

Enter Product Name: Notebook

Enter Quantity: 15

Enter Price: 50

Product added successfully!

Inventory Management System

1. Add Product
2. View Products
3. Update Product
4. Delete Product
5. Search Product
6. Exit

Enter your choice: 2

Inventory

ID: 101

Name: Notebook

Quantity: 15

Price: 50.0

Inventory Management System

1. Add Product

2. View Products

3. Update Product

4. Delete Product

5. Search Product

6. Exit

Enter your choice: 5

Enter Product ID to search: 101

Product Found:

Name: Notebook

Quantity: 15

Price: 50.0

Inventory Management System

1. Add Product

2. View Products

3. Update Product

4. Delete Product

5. Search Produ

6. Exit

CONCLUSION

The Mini Inventory Management System successfully demonstrates how basic software can be used to automate and simplify inventory-related tasks in small-scale businesses and organizations. The system replaces traditional manual methods with a digital approach, thereby reducing human errors, saving time, and improving the accuracy of data management.

Through this project, essential functionalities such as adding, updating, deleting, and searching product records are effectively implemented. The use of file handling ensures that data is stored permanently, making the system more reliable and practical for real-world applications. Additionally, features like stock tracking help users monitor inventory levels and make better decisions regarding restocking.

The system is designed with simplicity and user-friendliness in mind, making it easy to operate even for users with minimal technical knowledge. It also provides a strong foundation that can be further enhanced by integrating advanced features such as graphical user interfaces, database connectivity, and report generation. Overall, the Mini Inventory Management System improves efficiency, organization, and productivity in inventory handling, and it highlights the importance of automation in modern business operations.

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